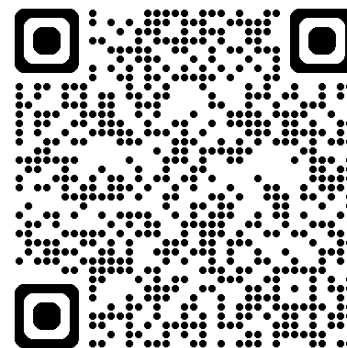




深度學習 Deep Learning

圖神經網路
Graph Neural Network (GNN)

Instructor: 林英嘉 (Ying-Jia Lin)
2025/12/03



[Course GitHub](#)



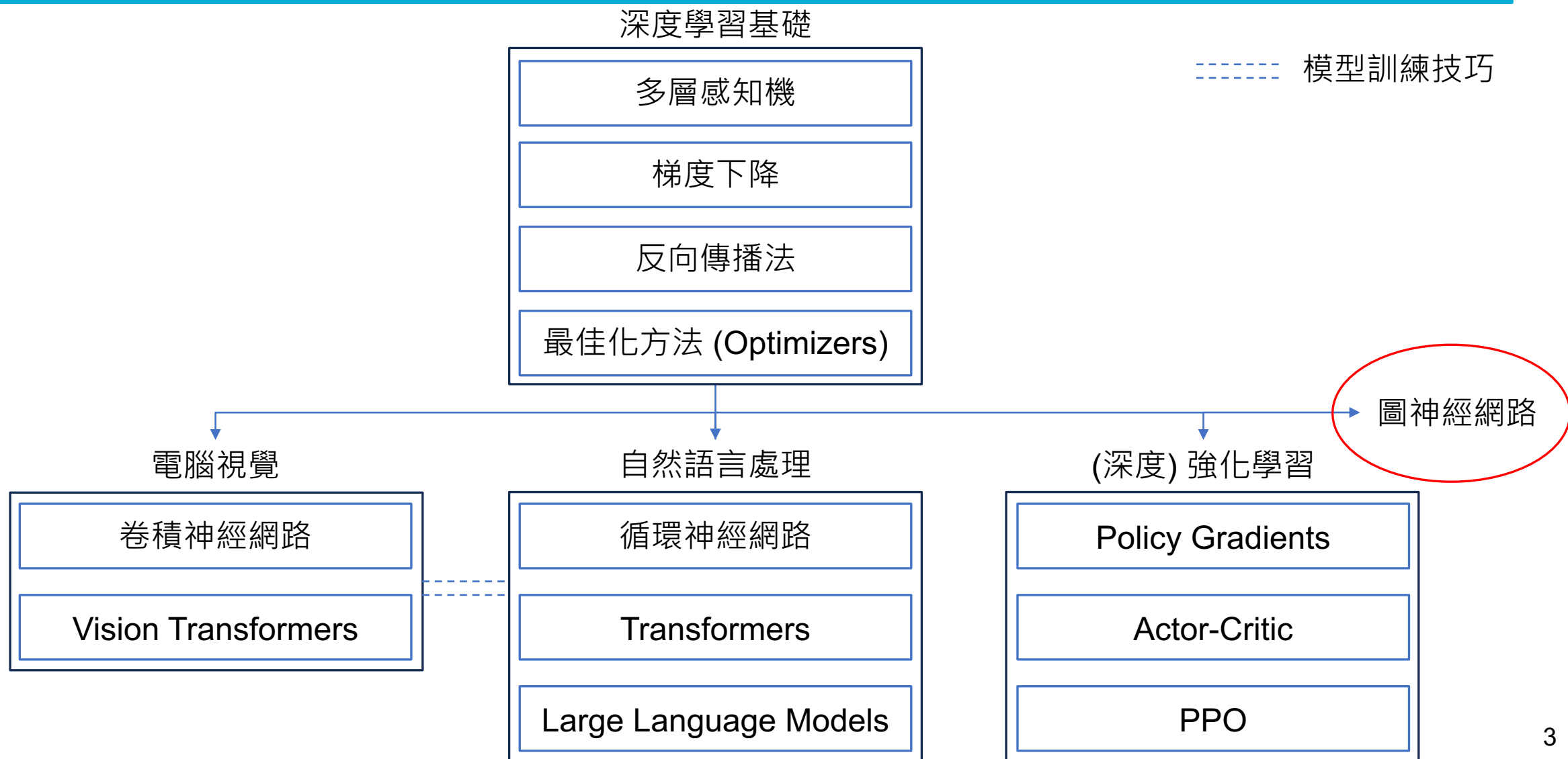
[Slido # DL1203](#)

Outline

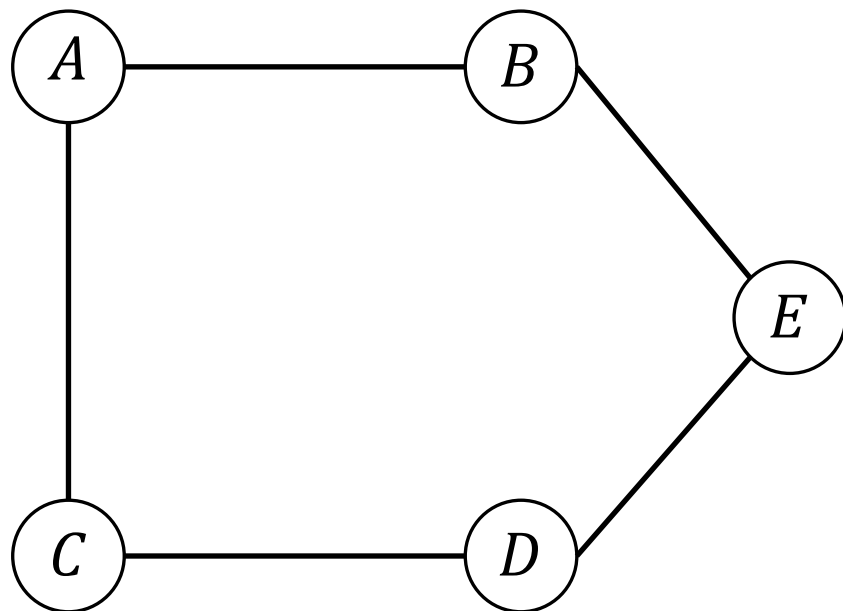
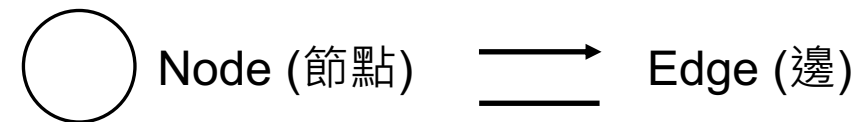
- 圖資料結構介紹 – 15 min
- GCN (Graph Convolutional Network) – 20 min
- GAT (Graph Attention Network) – 20 min
- Code – 30 min
- Project presentations – 40 min
- Quiz – 15 min



深度學習課程地圖

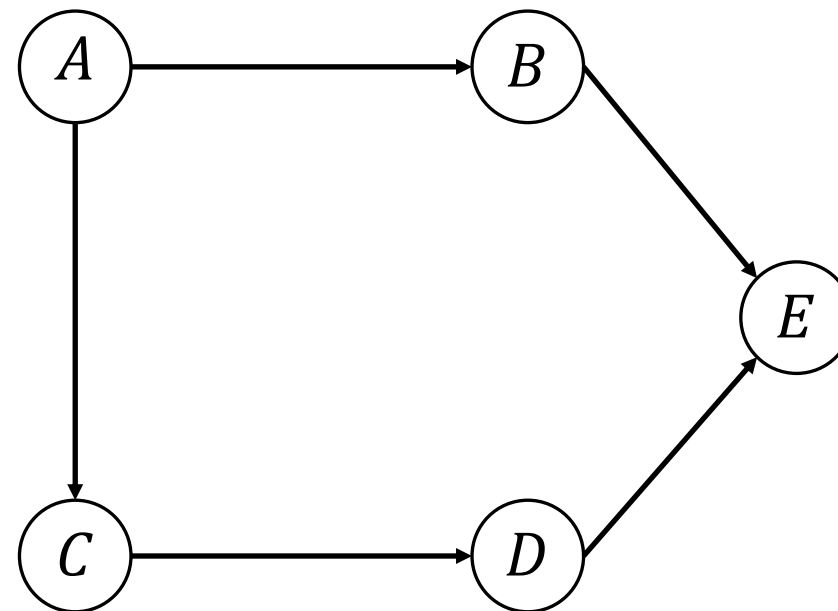


Graphs



cyclic

Undirected graph (無向圖)



acyclic

Directed graph (有向圖)



Graph Example (推薦系統)

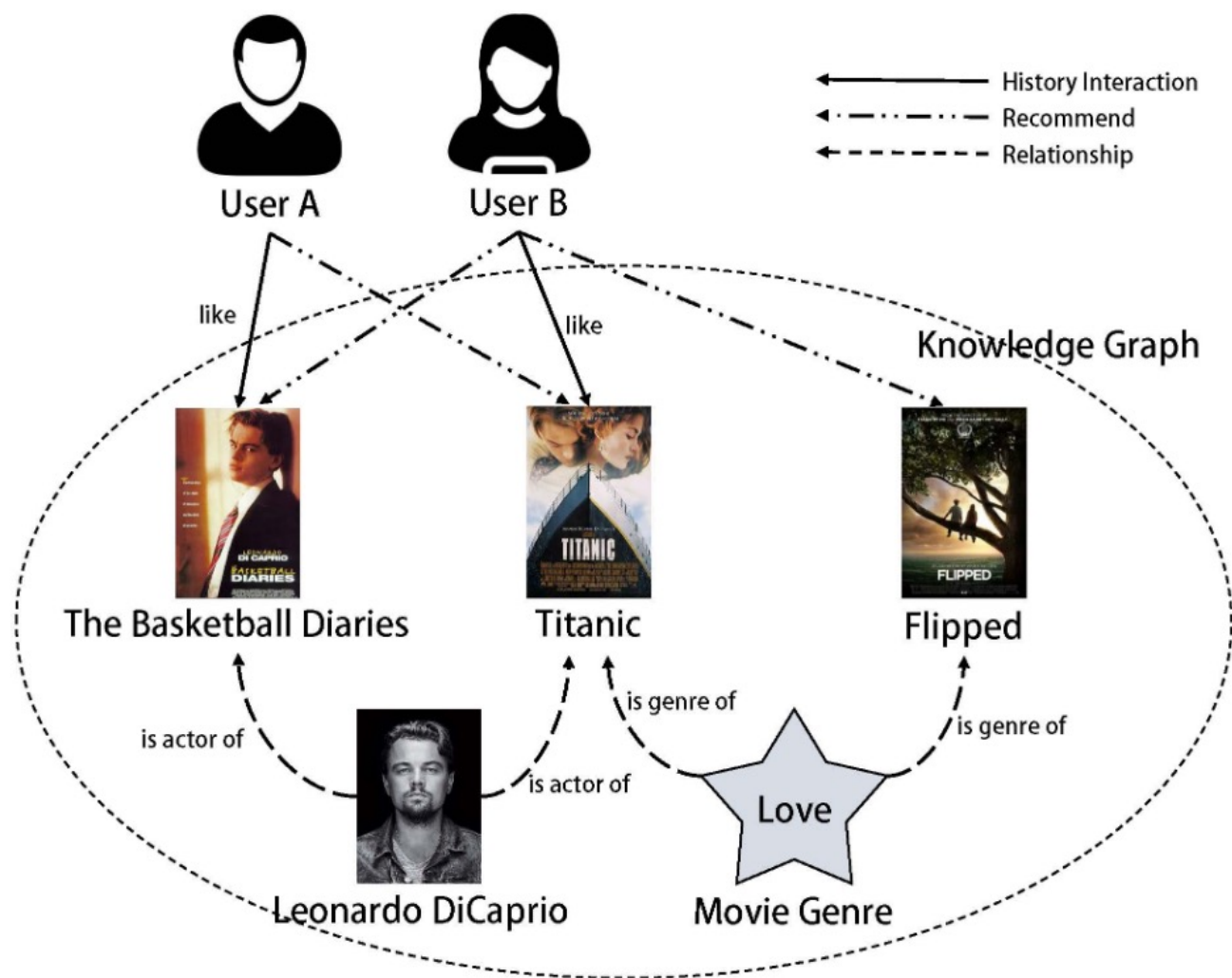


Figure source: Huang, Xiaoli, Junjie Wang, and Junying Cui. "A Personalized Collaborative Filtering Recommendation System Based on Bi-Graph Embedding and Causal Reasoning." Entropy 26.5 (2024): 371.



Graph Example (基因關聯)

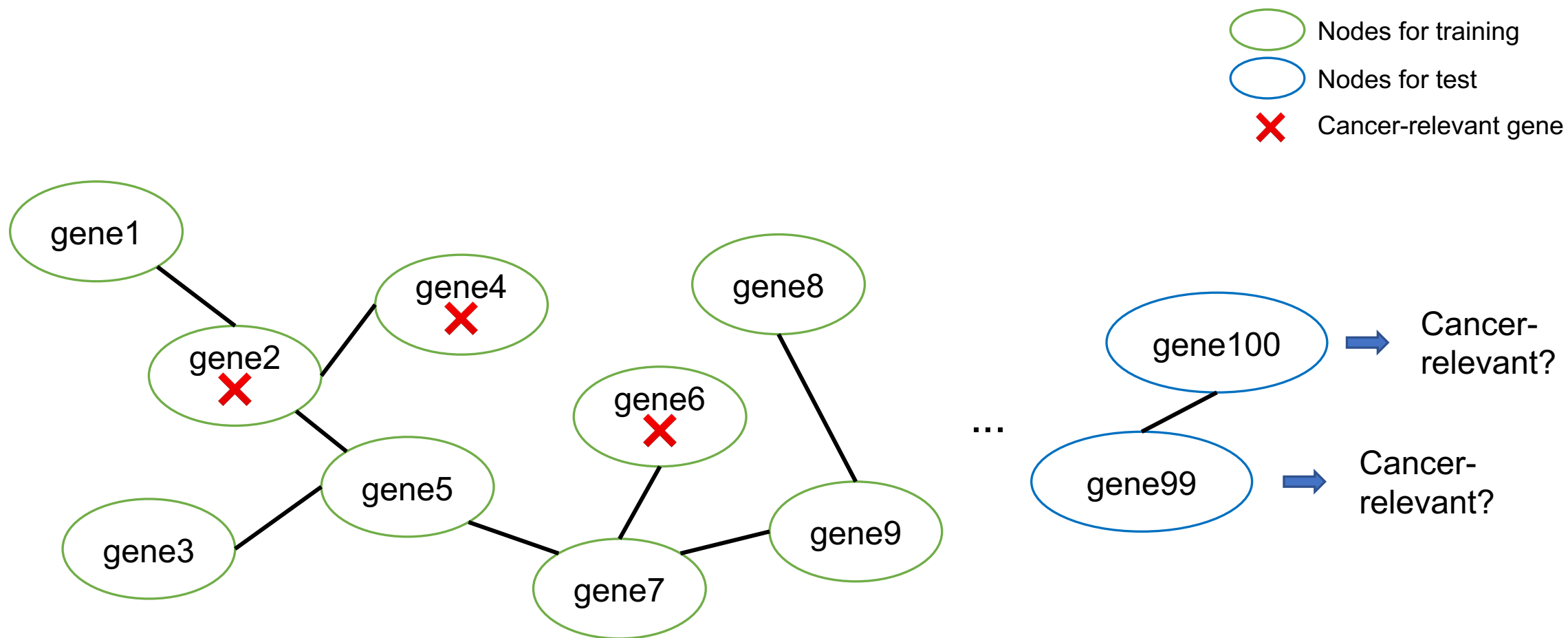
Figure source: Zhou, Cong-Ya, et al. "Evaluation of the target genes of arsenic trioxide in pancreatic cancer by bioinformatics analysis." *Oncology Letters* 18.5 (2019): 5163-5172.



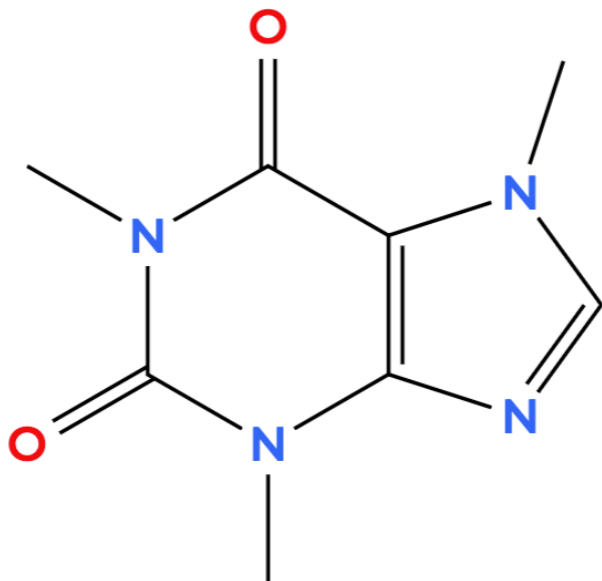
Figure 2. - A visual display of Kyoto Encyclopedia of Genes and Genomes pathways in 6 arsenic trioxide target genes (AKT1, CCND1, CDKN2A, IKBKB, MAPK1 and MAPK3) associated with pancreatic cancer.



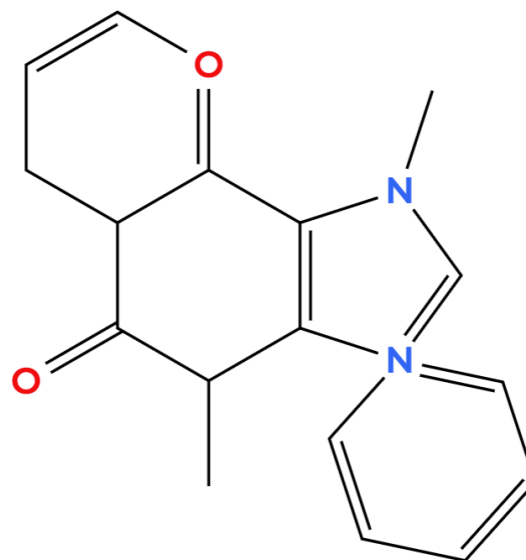
圖常見任務 (1) : Node Classification



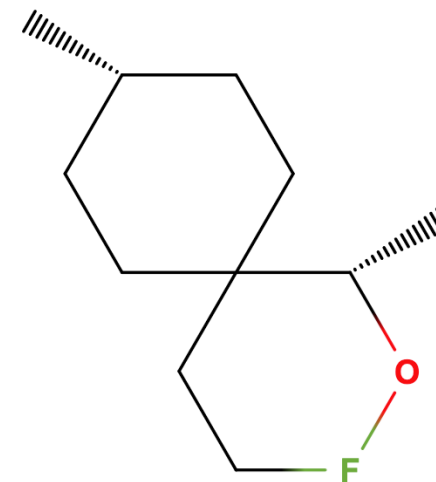
圖常見任務 (2) : Graph Classification



Toxic



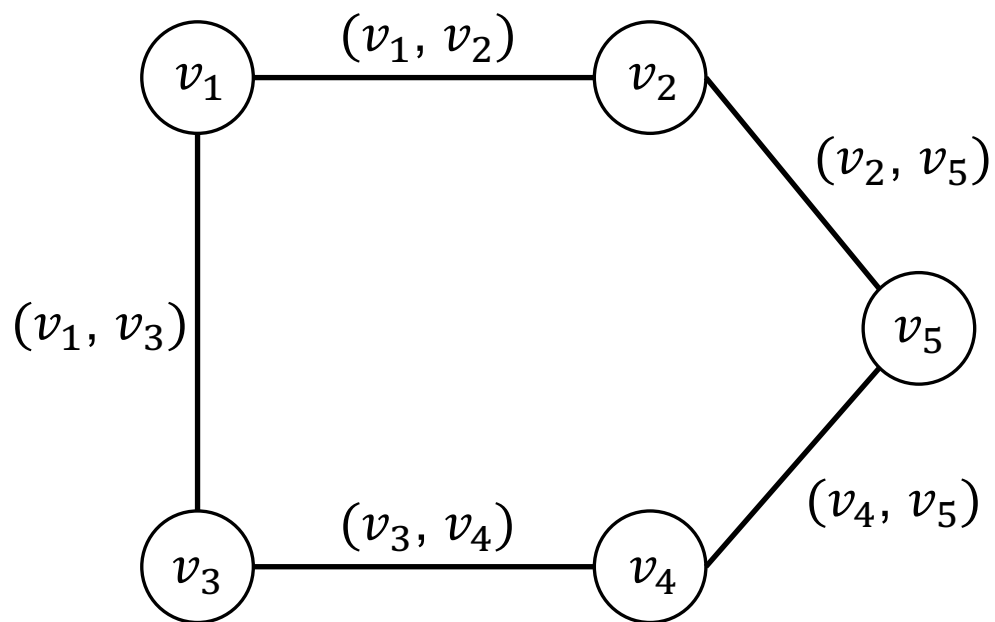
Not Toxic



?



Graph Terminology



Undirected graph (無向圖)

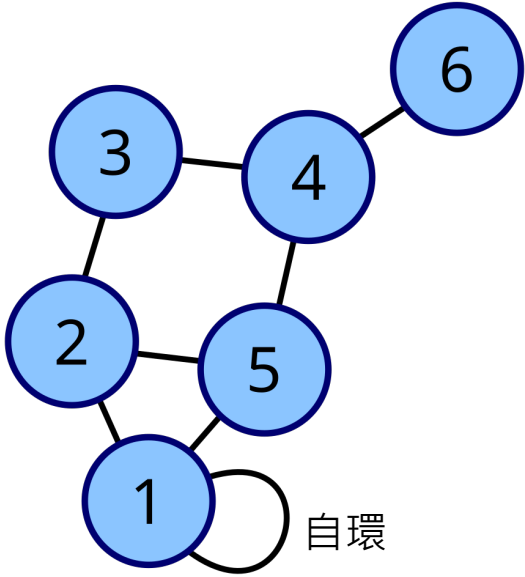
一張圖 G 寫作 $G = (V, E)$,

其中 :

- $V = \{v_1, v_2, \dots, v_n\}$
- $E \subseteq V \times V$, 如 (v_i, v_j) , (i, j) 為 G 中任意相鄰的兩個節點 (node)
 - i 通常代表起點 ;
 - j 通常代表終點



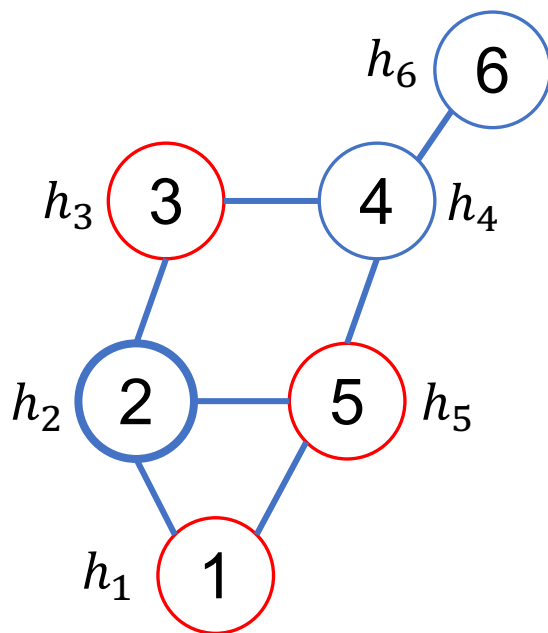
Adjacency Matrix and Degree Matrix

Vertex labeled graph	Degree matrix (度矩陣)	Adjacency matrix (鄰接矩陣)
 自環	$D = \begin{pmatrix} 4 & 0 & 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$ 對角矩陣	$A = \begin{pmatrix} 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{pmatrix}$



GNN 的核心步驟 (1)：聚合 (Aggregate)

💡 Aggregate: 收集鄰居資訊



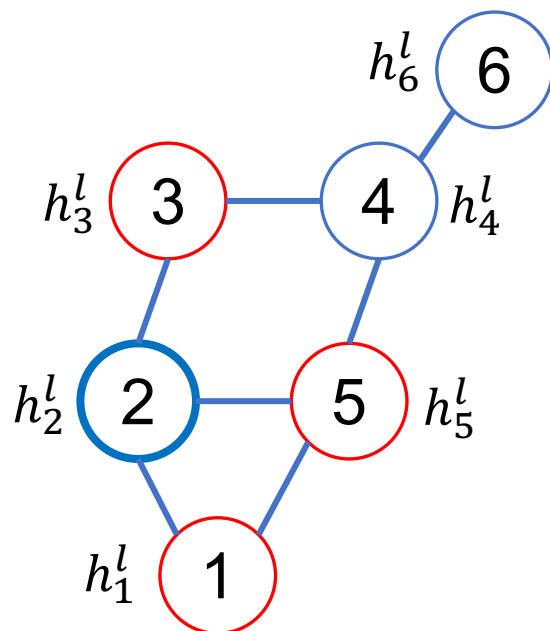
$$m_v = \text{AGGREGATE}(\{h_u | u \in \mathcal{N}(v)\})$$



h_v : 節點 v 的特徵表示

GNN 的核心步驟 (1)：聚合 (Aggregate)

💡 Aggregate: 收集鄰居資訊

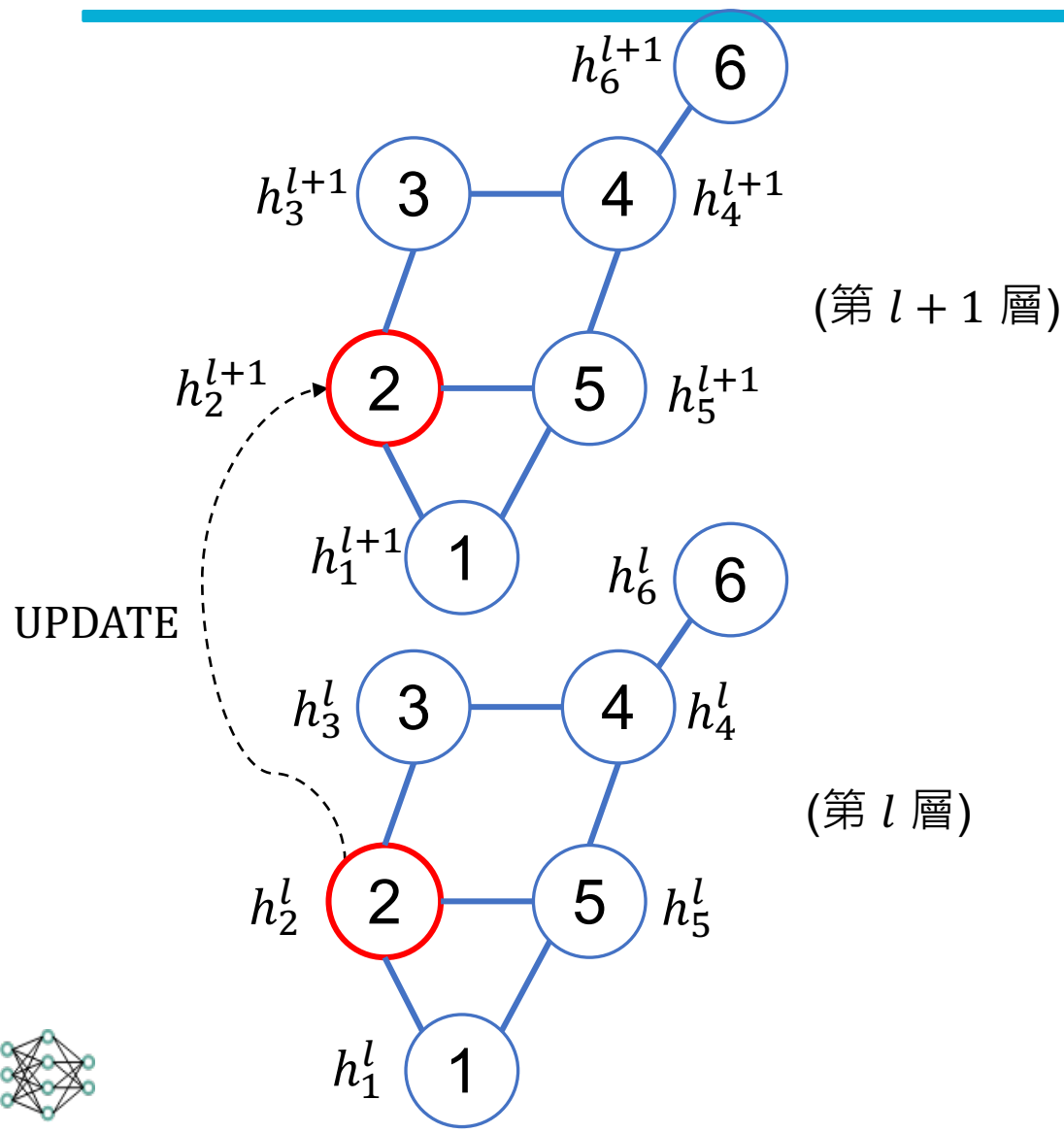


$$m_v^l = \text{AGGREGATE}(\{h_u^l | u \in \mathcal{N}(v)\})$$



h_v^l : 第 l 層中，節點 v 的特徵表示

GNN 的核心步驟 (2)：更新 (Update)



💡 Combine: 收集到鄰居訊息後，節點會根據這些資訊更新自己的特徵表示

$$h_v^{l+1} = \text{UPDATE}(h_v^l, m_v^l)$$

h_v^l : 第 l 層中，節點 v 的特徵表示

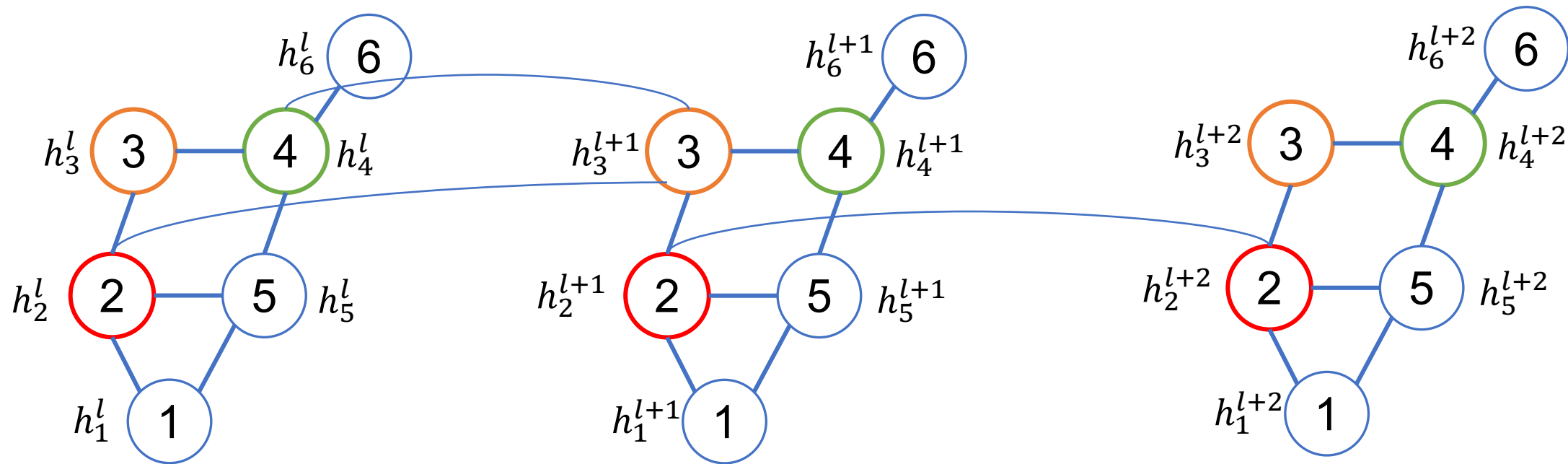


Message Passing

💡 一個節點的特徵，應該由它自身的特徵以及它所有鄰居的特徵共同決定。透過多層 (Layers) 的訊息傳遞，節點可以獲取更遠距離的資訊



Message Passing 說明範例 (1)



(第 l 層)

(第 $l+1$ 層)

(第 $l+2$ 層)

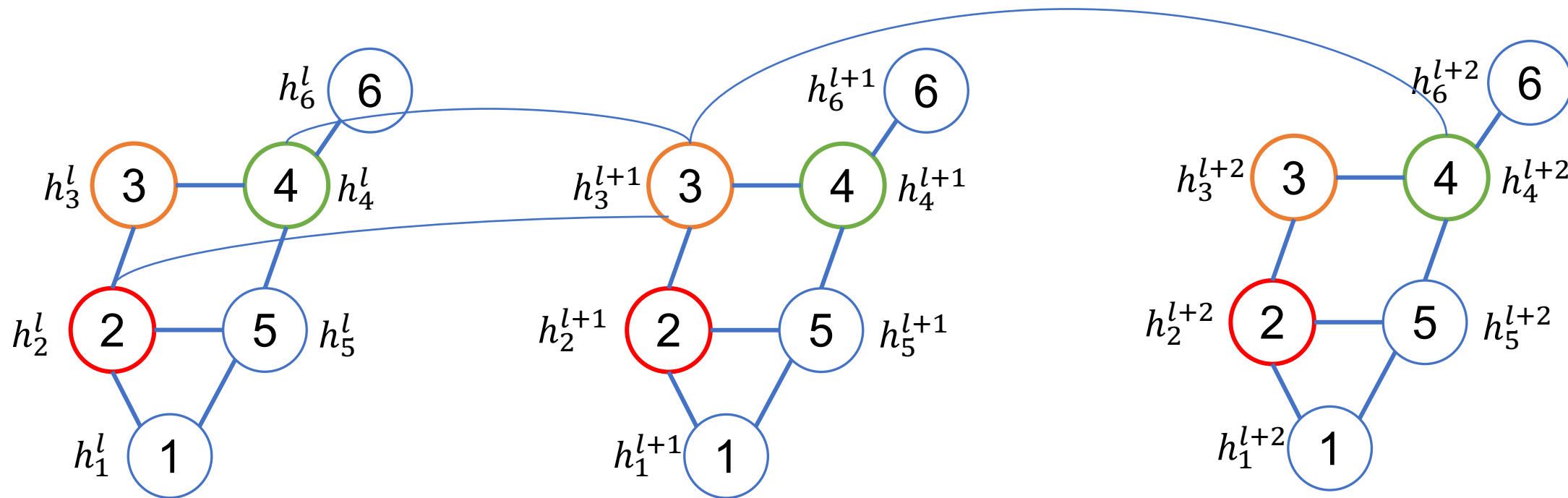
2號node無法接觸
到4號node

3號node帶有4號
node的資訊

2號node帶有4號
node的資訊



Message Passing 說明範例 (2)



(第 l 層)

(第 $l+1$ 層)

(第 $l+2$ 層)

2號node無法接觸
到4號node

3號node帶有4號
node的資訊

4號node帶有2號
node的資訊



Graph Convolutional Networks (GCN)

Thomas N. Kipf, , Max Welling. "Semi-Supervised Classification with Graph Convolutional Networks." International Conference on Learning Representations. 2017.

(Recap) GNN 核心步驟

以 Node classification 為例

Step1: 聚合鄰居節點資訊 AGGREGATE

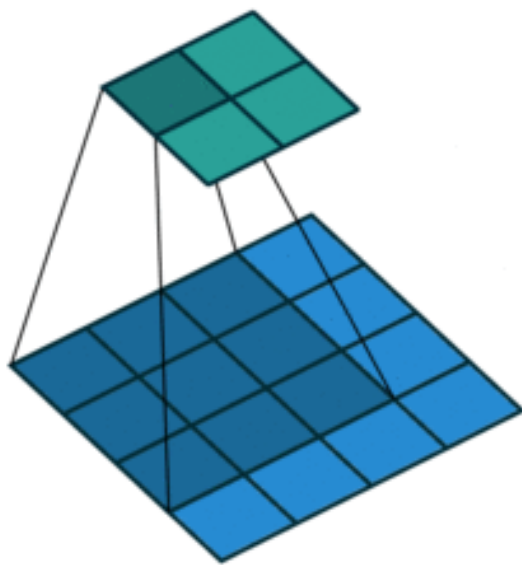


Step2: 更新自身節點資訊 UPDATE

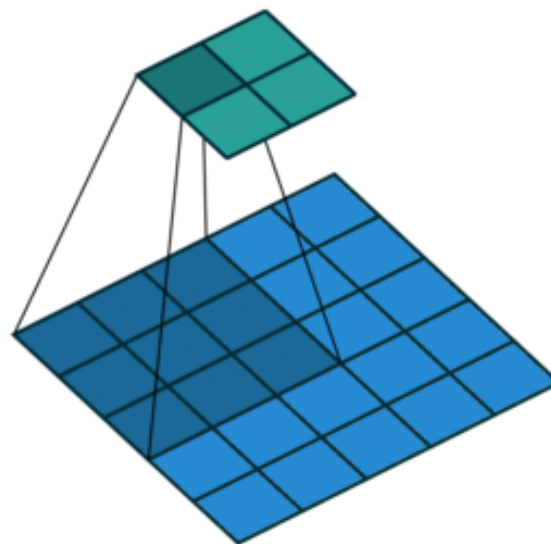


(Recap) Examples of Convolutions

Padding = 0, stride = 1



Padding = 0, stride = 2



💡 卷積 (convolution) 就是讓一個「小的矩陣」(filter) 在影像上滑動，進行局部加權平均 (局部區域的內積)，產生新的影像 (特徵圖，feature map)

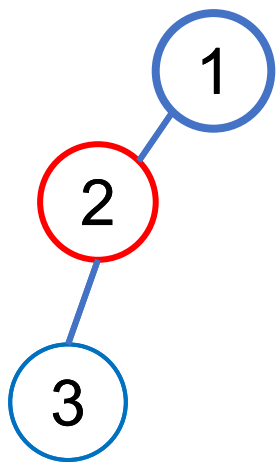
Figure source:

<https://hannibunny.github.io/mlbook/neuralnetworks/convolutionDemos.html>



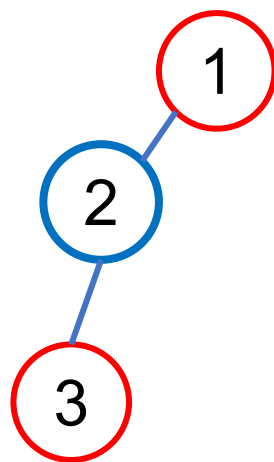
GCN 會先對鄰居收集資訊

💡 Aggregate: 收集鄰居資訊



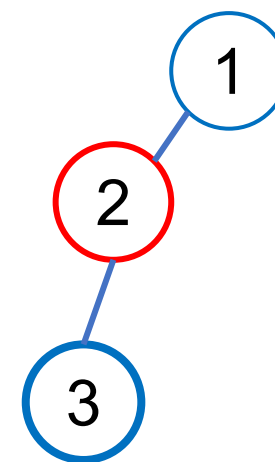
Source node: 1

Neighbor node: 2



Source node: 2

Neighbor node: 1, 3



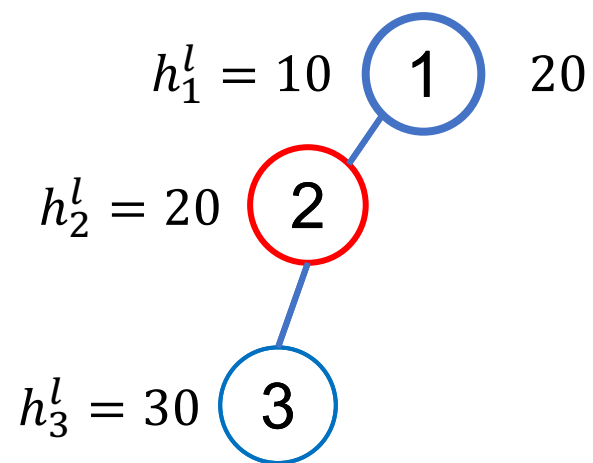
Source node: 3

Neighbor node: 2



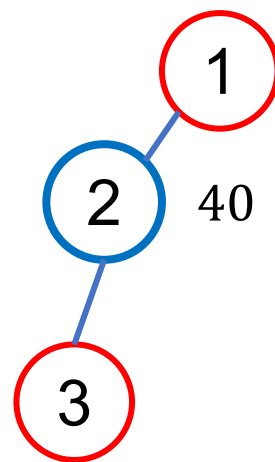
GCN 如何做 AGGREGATE ?

💡 GCN 使用相加來收集鄰居資訊



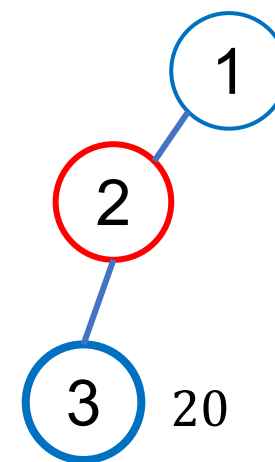
Source node: 1

Neighbor node: 2



Source node: 2

Neighbor node: 1, 3



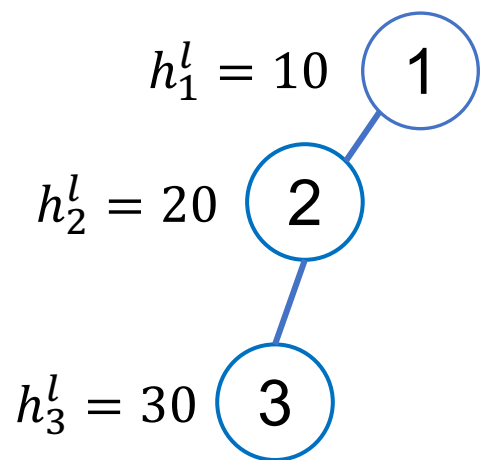
Source node: 3

Neighbor node: 2



⚠️ 此頁slide為求簡潔，每個node僅使用 1 個feature

實作上 GCN 如何做 AGGREGATE ?



$$H = \begin{pmatrix} 10 \\ 20 \\ 30 \end{pmatrix}$$

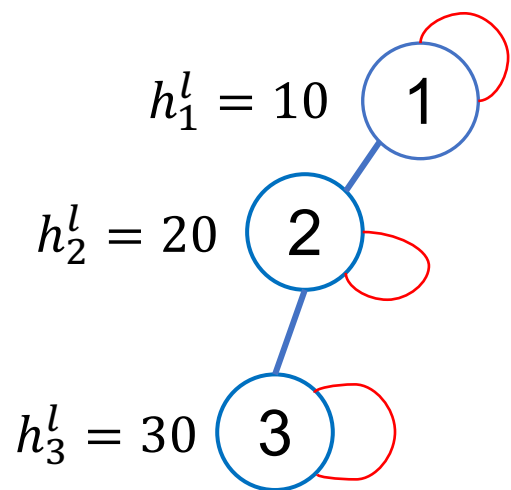
Adjacency matrix
(鄰接矩陣)

$$\tilde{A} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$\tilde{A}H = \begin{pmatrix} 20 \\ 40 \\ 20 \end{pmatrix}$$



實作上 GCN 如何做 AGGREGATE ?



$$\tilde{A} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \xrightarrow{\text{加入自環}} \tilde{A} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

💡 不僅要看鄰居，也要保留自己的資訊。



(Recap) GNN 核心步驟

以 Node classification 為例

Step1: 聚合鄰居節點資訊 AGGREGATE



Step2: 更新自身節點資訊 UPDATE



GCN 的 weights (UPDATE)

假設有一張圖的 nodes 有3個，且每個 node 的 feature 數目為10個

GCN_layer1 $H^{(2)} = \tilde{A} H^{(1)} W$

$\begin{matrix} \uparrow & \uparrow & \uparrow \\ 3 \times 3 & 3 \times 10 & 10 \times \text{hidden_size} \end{matrix}$

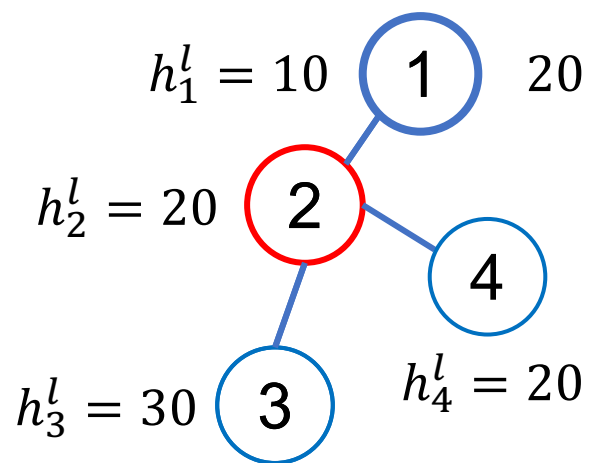
GCN_layer2 $H^{(3)} = \tilde{A} H^{(2)} W$ $\Rightarrow$ **Node Classification**

$\begin{matrix} \uparrow & \uparrow & \uparrow \\ 3 \times 3 & 3 \times \text{hidden_size} & \text{hidden_size} \times \text{num_classes} \end{matrix}$



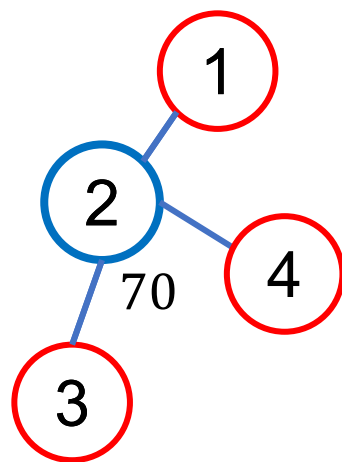
可能有些 node(s) 的鄰居特別多

💡 GCN 使用相加來收集鄰居資訊



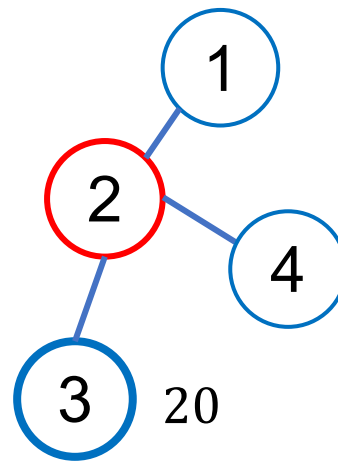
Source node: 1

Neighbor node: 2



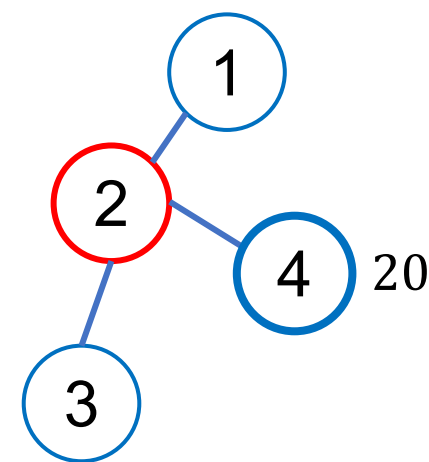
Source node: 2

Neighbor node: 1, 3, 4



Source node: 3

Neighbor node: 2



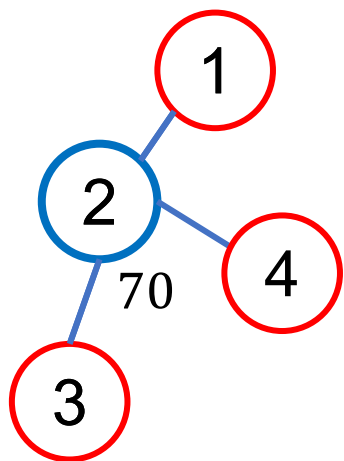
Source node: 4

Neighbor node: 2



⚠️ 此頁slide為求簡潔，每個node僅使用 1 個feature

GCN 對鄰接矩陣做標準化



Source node (i): 2

Neighbor node (j): 1, 3, 4

$$\tilde{A}_{ij} \times \frac{1}{\sqrt{\text{degree}(i)}\sqrt{\text{degree}(j)}} H \quad \leftarrow \text{標準化}$$

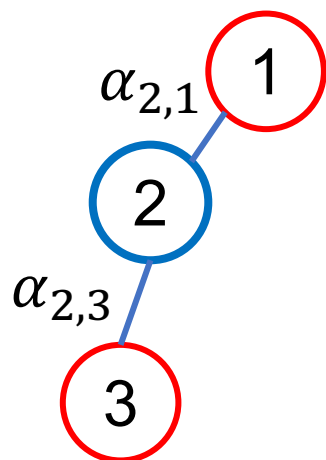


Graph Attention Networks (GAT)

Petar Veličković, , Guillem Cucurull, Arantxa Casanova, Adriana Romero, Pietro Liò, Yoshua Bengio. "Graph Attention Networks." International Conference on Learning Representations. 2018.

GAT 核心概念

$\alpha_{i,j}$ 代表 node j 對 node i 的重要性



node1 對 node2 的影響有多大？

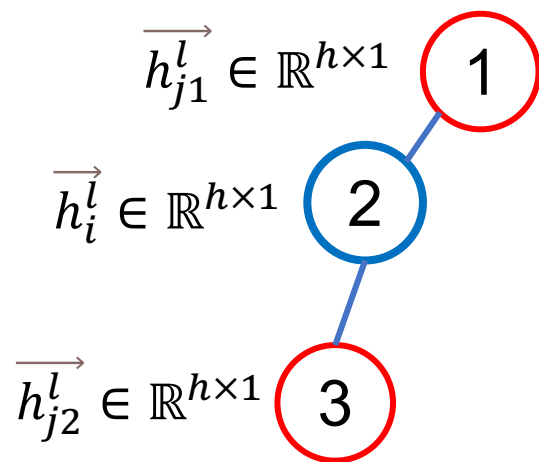
node3 對 node2 的影響有多大？

GCN 直接用相加來計算鄰居的重要性 ➡
🤔 重要性可不可以讓模型自己學？



Attention Mechanism in GAT (1)

Step1: AGGREGATE



每個節點都會經過線性
轉換 W ($W \in \mathbb{R}^{h \times h}$)

$$\vec{h}_{j1}^l W$$

$$\vec{h}_i^l W$$

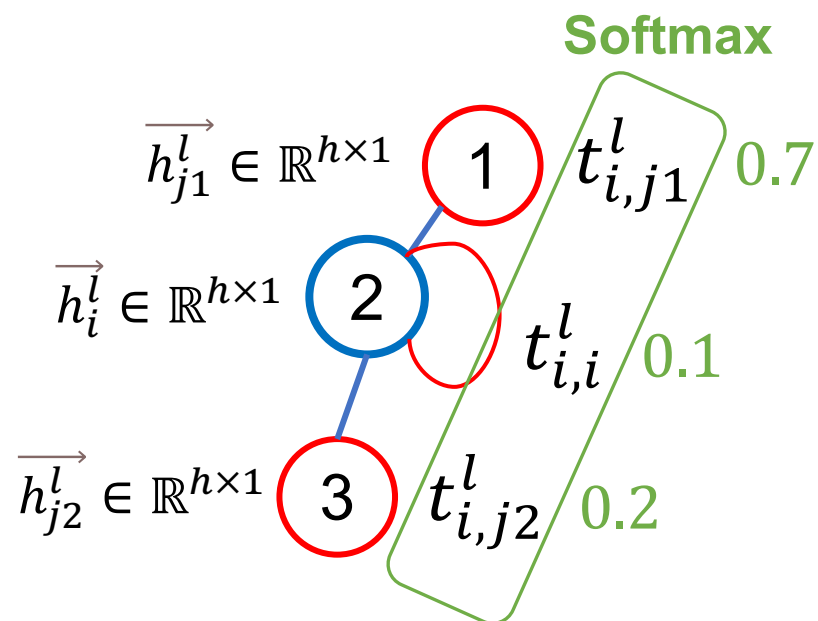
$$\vec{h}_{j2}^l W$$



Attention Mechanism in GAT (2)

Step1: AGGREGATE

假設我們正在算 j 對 i 的重要性



$$\begin{bmatrix} \text{blue} & \text{blue} & \text{blue} & \text{orange} & \text{orange} & \text{orange} \end{bmatrix} \times \begin{bmatrix} \text{yellow} \\ \text{yellow} \\ \text{yellow} \\ \text{yellow} \\ \text{yellow} \\ \text{yellow} \end{bmatrix} = t_{i,j}^l$$

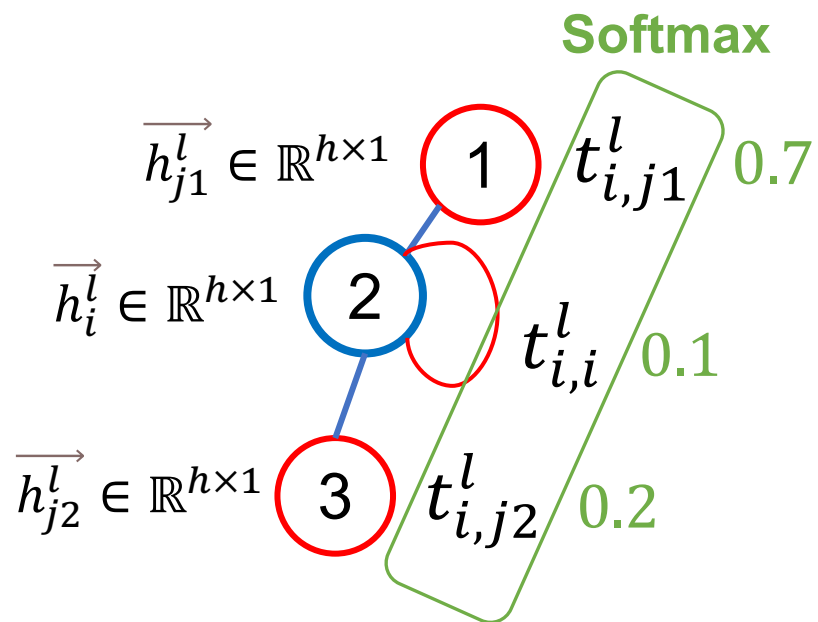
The first vector is labeled $\vec{h}_i^l W$ and the second vector is labeled $\vec{h}_j^l W$. The result is $t_{i,j}^l$.

讓模型自己去學習哪些
鄰居比較重要 (attention)



GAT 更新節點狀態的方式 (UPDATE)

Step2: UPDATE



$$\vec{h}_i^{l+1} = 0.7 * \vec{h}_{j_1}^l W + 0.1 * \vec{h}_i^l W + 0.2 * \vec{h}_{j_2}^l W$$



圖任務 Summary

假設有一張圖的 nodes 有3個，且每個 node 的 feature 數目為10個

GCN_layer2 $H^{(3)} = \tilde{A}H^{(2)}W$ $\Rightarrow$ **Node Classification**

3×3 $3 \times \text{hidden_size}$ $\text{hidden_size} \times \text{num_classes}$

READOUT $h^{(G)} = \text{MEAN}(H^{(3)})$ $\Rightarrow$ **Graph Classification**



延伸學習

- 李宏毅老師助教課
 - <https://youtu.be/eybCCtNKwzA?si=bCBjHODKbrxGiGlz>
 - <https://youtu.be/M9ht8vsVEw8?si=qYy2ggGluF8IEsE->
- 友善網誌
 - <https://jacky-18008.medium.com/圖神經網路怎麼這麼棒-how-powerful-are-graph-neural-networks-42bbfd640e68>
 - <https://medium.com/data-science/graph-convolutional-networks-introduction-to-gnns-24b3f60d6c95>
- 王佑中老師 slide
 - https://docs.google.com/presentation/d/1Xt_DThN7dd7w5Oay6RikgUhPx5wWw24xTQp7fQdEznA/edit?usp=sharing



Thank you!

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